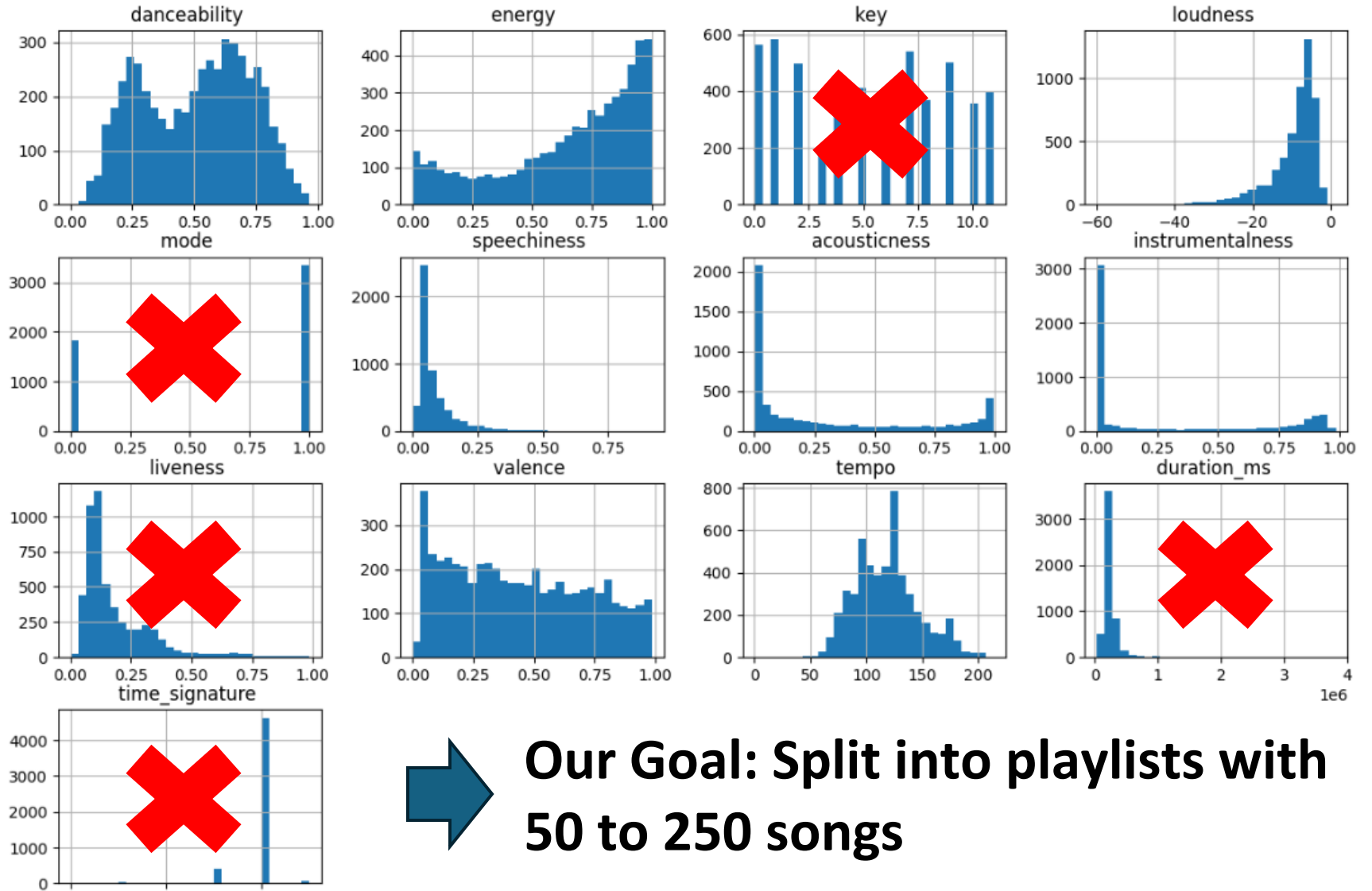


Unsupervised Machine Learning & Playlist Generation Results

Case study: Moosic

The Dataset & Goal

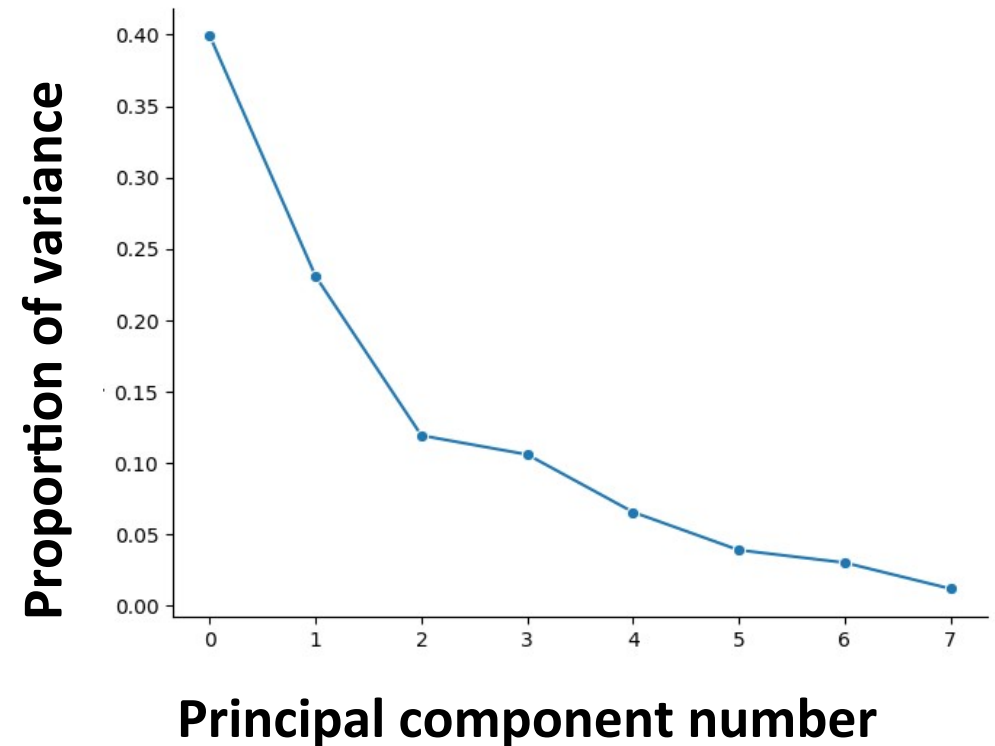
- The Dataset: 5235 songs
- Audio Features: 8 used / 5 excluded



Our Goal: Split into playlists with 50 to 250 songs

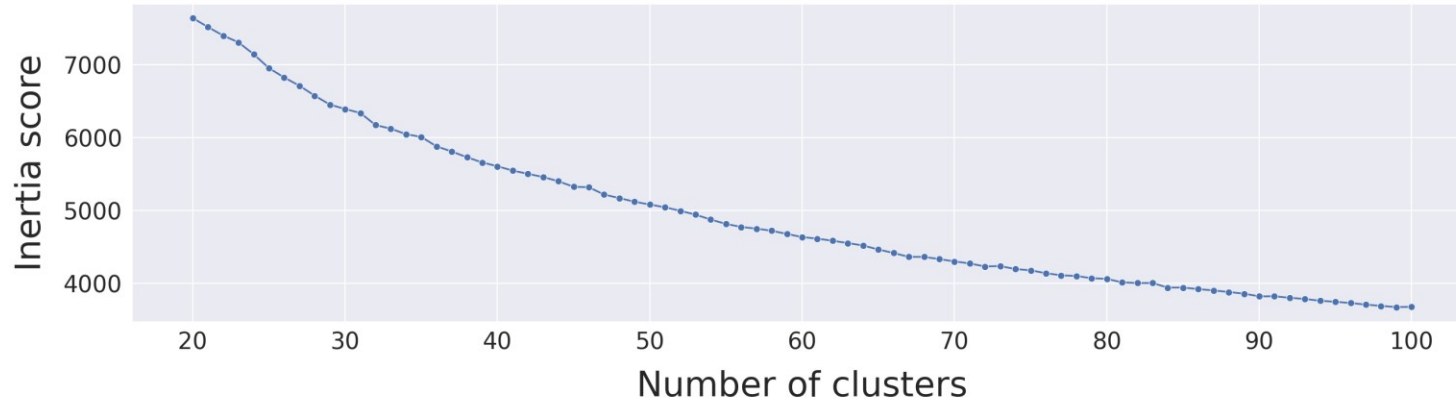
The Clustering Process

- K-Means method
- Standard Scaler
 - Allows data to spread out, not "squashing" distances for features with lower variation. The model can "see" that a subtle variance in speechiness is just as important as a massive variance in energy.
- Principal Component Analysis

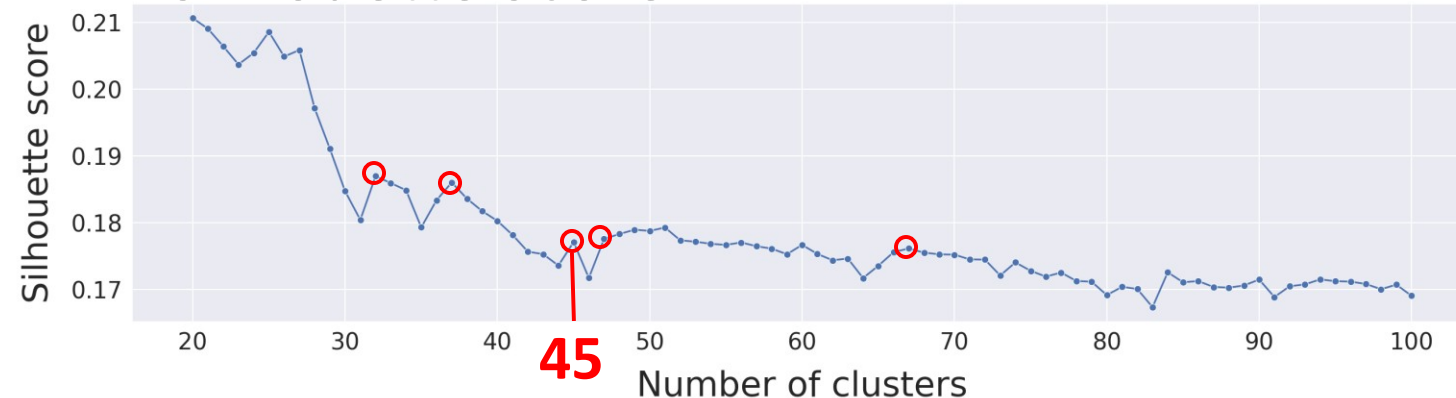


The Clustering Process

Inertia Score



Silhouette Score



- Candidates Considered based on silhouette score: Clusters 32, 36, 45, 47, and 67

- Decision Factor: Business requirements (50–250 songs per playlist)

- Final Choice (k=45):** Only 5 outliers fell outside the target range

Results

song_id	artist	name
3825	Napalm Death	Mind Snare
4011	Sacrificium	Death to Death
4050	Benediction	Certified...?
368	Alvays	Archie, Marry Me
3417	Massacre	Dawn of Eternity
4153	Polluted Inheritance	Faces
3119	Fresh	Signal
4026	Gorguts	A Path Beyond Premonition
4310	Morta Skuld	Devoured Fears
4191	Nile	Divine Intent

Playlists show a trend of clustering similar songs, but when looking more carefully into them, songs that don't really match the general genre or mood of the playlist come up.

song_id	artist	name
2002	Hauschka	Curious
5226	Ottorino Respighi	Pines Of Rome, P. 141: 3. The Pines Of The Janiculum
4519	Kara-Lis Coverdale	Moments In Love
1961	Pieter de Graaf	Silver Lining
2045	Ludwig van Beethoven	Piano Sonata No. 14 (quasi una Fantasia) in C-Sharp Minor, Op. 27 No.2 "Moonlight": I. Adagio sostenuto
4572	Loscil	Angle of List
2131	Henry Smith	A Tree & A River
2100	Johann Sebastian Bach	Stanza BWV 855a "The Children's Room" - Re-Recorded Version
4540	Stars Of The Lid	Don't Bother They're Here
4523	Hiroataka Shirotsubaki	Yanagiwara

Conclusions

- **Presented audio features not fully capable** of identifying similar songs as defined by human criteria.
 - Some defining **audio features might be very similar** for songs of different genres.
 - A **genre column** might help in identifying similar songs more clearly.
- **K-Means not the best method** for the case study
 - **Easy and fast** way to handle data.
 - **Might overlook nuances** that distinguish the different songs if not enough features are available.

Future Directions

- Applying weights to the features to highlight the most important ones.
- Using other features (genre) or creating custom indices from the ones available (aggression index, based on energy, valence and danceability).
- Adjusting playlists based on real user interactions and skips.
- Trying a different clustering method than K-Means.